

Inbred Animal_Yuki Wu

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by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi Wu
Date	20/11/2023
Module	Genetics
Coursework item	Essay

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?

Yes, in the previous essay writing assignment, I did not manage my time properly. This time I changed my plans and forced myself to complete it a few days before.

2. What do you feel went well? (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I did my best to include primary literature from the past 5 years in my essay.
I did a good job on time management in writing this essay, by separating tasks into several days to complete it, instead of finishing it by the last minute.
I read many literature and gained a lot of knowledge from this coursework.

3. What do you feel did not go well? (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

I think I could practice more on paraphrasing sentences, which I found hard during writing the essay.

4. What did you hope to learn from doing this coursework? Did you achieve that?

I hope that I can learn more about this essay topic. I did gain much information from this activity by reading literature on this topic.

The use of inbred animals in experimental setups in medical research

Introduction

A group of animals known as an inbred strain results from at least twenty successive generations of brother-sister that can be traced back to a single founding pair of individuals (Carter et al., 1952). The genetic variance can theoretically reach zero if enough generations are inbred (Wright, 1934). Bailey (1982) concluded that using twenty consecutive generations is minimal, as it will create approximately 200 polymorphic loci in the 20th generation. Therefore, a maximum level of 2% of the genetic variance has been set as a criterion when generating inbred strains.

Mice are one of the notable species for inbred strains and were widely used for scientific research in the twentieth century. The benefits of employing mouse models encompass clearly defined genomic data, established genetic manipulation and transgenesis protocols, a comparatively short lifespan and reproductive cycle, a small size that facilitates housing and maintenance, generally submissive, and the convenience of maintaining a genetically homogeneous lineage (Hickman et al., 2017; Tuttle et al., 2018). This essay explores the use of inbred animals in experimental setups in medical research, focusing on inbred mice.

v. good intro

Generation of inbred mice

To generate inbred mice, a controlled breeding process begins with selecting a pair exhibiting desired traits to serve as founders. The strain becomes highly homogeneous through successive sibling mating for over 20 generations. Individuals being >98% isogenic in the population can be achieved by inbreeding for more than 150 generations (Festing, 1981; Beck et al., 2000). The term *isogenic* describes the genotypic condition of a locus in which all individuals within an inbred generation are homozygous for the identical allele inherited from a common ancestor (Bailey, 1982).

Selective breeding, backcross mating, and continuous genetic monitoring are essential. Careful record-keeping and calculating the inbreeding coefficient help maintain the desired level of homozygosity, where the inbreeding coefficient measures the probability that two alleles at a randomly chosen locus are identical by descent (Wright, 1922). As the coefficient increases, so does the genetic stability. If DNA is required to be analysed, it can be retrieved using a typical salt extraction approach that includes an initial Proteinase K digestion phase from inbred mice tail tissue (Chebib et al., 2021).

do you mean genotyping?

Use of inbred mice in medical research

Inbred mice are used in various experiments that need genetically identical animals, and they are especially valuable in biomedical and behavioural research. In biomedical research, the laboratory mouse is the most frequently used experimental non-human subject (Hickman et al., 2017). For decades, inbred mouse strains have been preferred over outbred populations, with specific strains such as C57BL/6 and BAL/c (Festing, 2014).

In medical research, they serve as models for human diseases, studying specific disease pathways and exploring treatments or cures (Tuttle et al., 2018). For example, in cancer research, Dabbagh Moghaddam et al. used BALB/c inbred mice to test the anti-cancer effect of melittin-loaded niosome for breast cancer treatment. By using 40 BALB/c inbred mice and estimating the renal and liver enzyme activity, they successfully concluded that melittin-loaded niosome is a more effective treatment against cancer than free melittin (Dabbagh Moghaddam et al., 2021). *good but why this example in particular? it's one out of millions.*

Moreover, in genetics studies, inbred mice enable the identification of specific genes responsible for traits or susceptibilities. For example, genetics studies on inbred mice offer valuable insights into identifying cocaine addiction genes due to the ease of controlling the environment and behavioural outcomes. *Trpv2*, an ionotropic cannabinoid receptor, is found to be the critical locus for cocaine self-administration, where the behaviour data and brain RNA-seq data were obtained from 41 inbred mice strains (Khan et al., 2023). In another investigation, 39 strains of BXD recombinant inbred mice were assessed for cocaine self-administration. The study identified a quantitative trait locus (QTL) related to cocaine self-administration that contained a trans expression QTL (eQTL) for *Fam53b*, which is the gene in which cocaine-use disorder variants have been identified (Dickson et al., 2016; Gelernter et al., 2014). *good*

Behavioural traits such as impulsivity can serve as effective substitutes for genetically mapping cocaine use disorders in rodent models (Kalueff et al., 2015). For instance, research with BXD recombinant inbred mice revealed an association between poor reversal learning, indicating a lack of inhibitory control, and increased cocaine self-administration (Cervantes, Laughlin & Jentsch, 2013). This suggests that inbred mice also play a significant role in behavioural studies.

Furthermore, inbred mice are vital for understanding diseases and developing treatments in immunology. With more than 90% genetic homology to humans, mice have sophisticated biological systems similar to humans, such as the immune systems. Research involving mice has contributed to identifying major histocompatibility complex genes and the T cell receptor. Additionally, it has advanced our comprehension of antibody synthesis regulation and various aspects of the immune system (Khanna & Burrows, 2011). Their use allows the manipulation of immune cells and genetic

modifications, enabling efficient disease studies. In studying immune system influences, inbred mouse models with controlled environments are employed. Exploring responses in mice with diverse exposures may also unveil insights for improved vaccines and adjuvants (Hamilton et al., 2020).
Here there are some really good examples as some inbred strains are more susceptible to some immune related diseases than others.

Genetic stability of inbred mice

The main reason that inbred mice are so valuable for medical research is their genetic stability. Being genetically stable allows the experiment to be reproducible and enables comparison with laboratory results worldwide (Silver, 2001; Festing, 2019). The genetic stability of inbred mice is characterised by a high degree of homozygosity, meaning that the alleles at a particular gene locus are nearly identical between individuals.

However, the genetic stability and isogenicity of inbred mice will be decreased by mutation, potential contamination, and heterozygote selection, leading to genetic impurity or allogenicity. Genes constantly mutate in inbred populations. Casellas and Medrano (2008) proved that about 4.5% of the total phenotypic variability is due to the accumulation of mutations when inbreeding generation after generation. However, applying frozen embryo storage can somewhat decrease the mutation level. By limiting the number of generations per year, this storage is predicted to prevent genetic drift (Bailey, 1982).

In addition, inadvertent outcrossing is one of the sources of contamination. It might inevitably happen when mating individuals with different strains uncontrollably, leading to a range of heterozygosity depending on their genetic differences. Sometimes, mislabelling also happens in the laboratory if it is improperly operated. Contaminations can be identified by observing the different phenotypes or conducting a genome scan (Casellas, 2011). Involving more than one couple per generation might generate heterozygotes as well. The heterozygotes selection might encourage heterozygotes as they have better reproductive fitness than homozygotes and might slow down the progress of inbreeding. A consistent, long-term full-sib mating or backcrossing practice can reduce heterozygote selection to negligible levels over time. (Bailey, 1982).

v. good

Limitations of the utility of inbred mice

Despite their advantages, the use of inbred mice comes with certain limitations. Although humans and mouse models have many similar characteristics, they may not precisely imitate certain human physiology and disease elements. Key differences include size, metabolic rate, life history, diets, microbiomes, and pathogen exposures (Kleiber, 1961; Nguyen et al., 2015). The genetic, epigenetic, and environmental variations contribute to distinct biology and disease patterns in mice compared to humans. As a result, mice models that only capture one element of the disease may not fully

depict the disease's complexity in humans. Furthermore, various mouse strains may respond differently to experimental treatments or develop diseases at different rates, resulting in diversity in experimental outcomes.

Finally, while mice are considered less ethically demanding to utilise in the study than larger animals or humans, ethical considerations must still be considered when researching mice. As genetic diversity might be lost resulting from the inbreeding process, leading to reduced fitness and increased mice's susceptibility to diseases due to the accumulation of deleterious recessive alleles, this raises ethical concerns in research, particularly related to inbreeding depression, genetic manipulations, and limited behavioural enrichment. Researchers must transparently justify the selection of inbred models, aligning with replacement, reduction, and refinement principles.

very good

Conclusion

In conclusion, using inbred mice in medical research offers distinct advantages, providing genetic homogeneity, reproducibility, and suitability for diverse studies. These mice are crucial models for understanding disease pathways, exploring treatments, and identifying genes. However, their genetic stability comes with trade-offs, such as reduced diversity, potential health risks, ethical considerations, and some genetic heterogeneity inevitably exists. While their use has contributed significantly to understanding diseases, it is essential to recognise inherent differences between mice and humans. Despite the limitations, inbred mice remain valuable animal models, offering clear advantages compared to outbred populations exhibiting high variability. Researchers must transparently justify their use, following ethical guidelines to balance scientific objectives with humane treatment and examine the potential effects on genetic stability.

(1437 words)

no figures???? add figures for higher marks

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FINAL GRADE

GENERAL COMMENTS

75/100

PAGE 1

PAGE 2

Text Comment. and still is

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PAGE 3

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PAGE 4

Text Comment. Here there are some really good examples as some inbred strains are more susceptible to some immune related diseases than others.

Text Comment. v. good

PAGE 5

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PAGE 6

PAGE 7

INTRODUCTION (10%)

75 / 100

FAIL (25)	Difficult to identify the topic and context from introduction; missing thesis statement
3RD (45)	Topic identifiable but introduction shows weak grasp of context; unclear thesis statement
2B (55)	Introduction gives relevant background context; fairly clear thesis statement, but with flaws, e.g. background only weakly linked to thesis; doesn't introduce all the necessary concepts
2A (65)	Introduction gives relevant background context; clear thesis statement; relates background to that thesis
1ST (75)	Introduction shows thorough knowledge of context; states and justifies thesis in relation to literature; background all logically related to thesis
HIGH 1ST (100)	Introduction shows thorough knowledge of context; states and fully justifies thesis; ample background provided by relevant primary literature

STRUCTURE (20%)

75 / 100

FAIL (25)	Structure is so poor that it is unclear what is being communicated
3RD (45)	Disorganised and inadequate paragraphing; gaps and jumps in structure make it difficult to extract meaning; large amounts of irrelevant material unrelated to the thesis; essay is a literal (or figurative) list of bullet points
2B (55)	Fairly organised paragraphs that mostly address the thesis, but with flaws, e.g. disjointed organisation of information within and between paragraphs forces reader to work out links for themselves; conclusion vague or introduces new concepts
2A (65)	Paragraphs are all well organised, have topic sentences, and support the thesis; transitions within and between paragraphs are mostly clear; conclusion sums up the arguments and author's thesis
1ST (75)	Paragraphs are all well organised, have topic sentences, and support the thesis; transitions within and between paragraphs are clear; conclusion reinforces arguments and justifies the author's thesis
HIGH 1ST (100)	Paragraphs are all well organised, coherent, have topic sentences, and support the thesis; clear transitions and narrative structure throughout; conclusion concisely reinforces arguments and fully justifies the author's thesis

ARGUMENT (40%)

75 / 100

FAIL (25)	Arguments unclear; essay consists mostly of incorrect or irrelevant information; reveals a lack of comprehension of the essential concepts or of the task
3RD (45)	Arguments are superficial; major omissions; major errors of interpretation of the literature
2B (55)	Arguments are fairly clear but with flaws, e.g. argument insufficiently explained; minor omissions; minor errors of comprehension; minor misinterpretation of the literature; shallow or unfocussed criticism of sources
2A (65)	Arguments are specific and well developed; no significant errors; shows some critical, analytical, and synthetic interpretation of the literature
1ST (75)	Arguments are specific, well developed and defended with concrete examples; shows a high level of critical, analytical, and synthetic skill in evaluating the limitations of sources
HIGH 1ST (100)	Arguments are specific, well developed and defended with concrete examples; consistently critical, analytical, and synthetic in evaluating the limitations of sources; and is additionally original, novel, or insightful

REFERENCES (15%)

75 / 100

FAIL (25)	Fails to reference most sources of information
3RD (45)	Fails to reference at least some sources of information; some cited sources are not referenced (or vice versa); some sources inappropriate or misused
2B (55)	All sources referenced but with flaws, e.g. references poorly formatted; some sources tangential to topic; overreliance on too few sources or on secondary sources such as reviews or textbooks
2A (65)	All sources referenced using appropriate Harvard style with only minor formatting flaws; all sources appropriate
1ST (75)	All claims in text substantiated and referenced using appropriate Harvard style; all sources well chosen and of high quality; flawless formatting (consistent 'Sentence case' for titles, italic Latin names, etc.)
HIGH 1ST (100)	All claims in text substantiated and referenced using appropriate Harvard style; all sources well chosen and of high quality; flawless formatting (consistent 'Sentence case' for titles, italic Latin names, etc.)

ENGLISH (10%)

75 / 100

FAIL (25)	Consistently poor use of English causes almost complete breakdown in meaning
3RD (45)	Consistently poor use of English vocabulary, spelling, grammar, punctuation and register makes it difficult to extract meaning; cohesive devices used repetitively, excessively or not at all; simple sentence construction is mostly accurate but

errors in attempts at more sophisticated sentences; rambling, colloquial or repetitive

2B (55)	Poor use of English impacts on meaning; cohesive devices are limited; complex structures are attempted but limited in range; may show systemic errors in e.g. subject/verb agreement, use of articles, consistent and appropriate tense/voice/mood; contains unexplained abbreviations; some jarring vocabulary or awkward phrasing
2A (65)	Largely accurate use of English; any small errors do not impact on meaning; uses a range of cohesive devices and sophisticated structures
1ST (75)	Accurate use of English; uses a wide range of cohesive devices and sophisticated structures; uses a wide range of appropriate vocabulary
HIGH 1ST (100)	Precise, accurate, varied and sophisticated use of English; author's voice is clear

FORMATTING (5%)

75 / 100

FAIL (25)	Far too long or far too short
3RD (45)	Inappropriately formatted; significantly over- or under-long
2B (55)	Formatting largely good, but with flaws, e.g. inconsistent formatting of headings; somewhat over- or under-long; not concise; figures and tables (if present) have poor legends and/or not referred to in text
2A (65)	Formatting very good throughout; differences in formatting of elements (headings, etc.) are all semantically meaningful; keeps more-or-less to word-limit; figures and tables (if present) have good legends and are referred to in text
1ST (75)	Formatting excellent throughout; very consistently formatted e.g. with Word styles, LaTeX, etc., as appropriate; concise and keeps to word-limit
HIGH 1ST (100)	Formatting and presentation is publication-standard; elements are all the length they need to be and no concise and keeps to the word-limit